Pakistani Supermarket Price Comparison Pipeline

Data Engineering & Analytics Project

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FAST NUCES

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GitHub: github.com/ahmedbutt1056/pk-supermarket-price-comparison

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1. Executive Summary

This project implements a comprehensive data engineering pipeline that scrapes, cleans, matches, and analyzes product prices across five major Pakistani retail chains: Metro Cash & Carry, Al-Fatah, Naheed, Daraz, and Jalal Sons. The pipeline processes real data scraped from live websites, applies multi-pass cross-store product matching, and generates actionable price intelligence.

Key Results

Total Raw Products:	239,744
Processed Products:	237,285
Stores Scraped:	5
Cities Covered:	6
Cross-Store Match Groups:	681
Cross-Store Matched Rows:	7,014
Analysis Charts Generated:	14
Excel Analysis Reports:	12

Data per Store

Alfatah:	124,695 products across 5 cities
Daraz:	68,952 products across 3 cities
Jalalsons:	6,201 products across 3 cities
Metro:	19,256 products across 4 cities
Naheed:	20,640 products across 4 cities

2. Project Architecture

The project follows a modular pipeline architecture with clearly separated concerns:

- scrapers/ - Web scraping modules for each store (Metro, Al-Fatah, Naheed, Daraz, Jalal Sons)
- processing/ - Data cleaning, normalization, brand extraction, unit standardization
- matching/ - Multi-pass cross-store product matching (deterministic + fuzzy)
- analysis/ - Statistical analysis, price dispersion, leader dominance index, correlation analysis
- validation/ - Data quality checks and pipeline validation
- app.py - Interactive Streamlit dashboard for data exploration
- config.py - Central configuration (paths, settings, store definitions)
- utils/ - Shared utilities (logger, helpers, brand dictionaries)

Technology Stack

- Python 3.12.3 - Core language
- pandas + PyArrow - Data manipulation and Parquet I/O
- BeautifulSoup4 + lxml - HTML parsing for Naheed & Jalal Sons scrapers
- Requests + Session rotation - HTTP client with anti-bot measures
- RapidFuzz - Fuzzy string matching for product resolution
- Streamlit - Interactive web dashboard
- Matplotlib + Seaborn - Statistical visualization
- Git + GitHub - Version control and collaboration

3. Data Collection (Web Scraping)

All product data was scraped from real, live Pakistani e-commerce websites. No synthetic or hardcoded data was used. Each store required a custom scraping strategy due to different website architectures:

3.1 Metro Cash & Carry

API: JSON REST API at metro.pk/api. Scraped 213 product categories across 4 cities (Karachi, Lahore, Islamabad, Faisalabad). Each category paginated with 40 products per page. Total: 19,256 products.

3.2 Al-Fatah

API: Shopify-based JSON API at alfatah.com.pk/products.json. Paginated with 250 products per page. Replicated across 5 cities. Total: 124,695 products.

3.3 Naheed

Method: HTML scraping with BeautifulSoup. Scraped 92 categories from naheed.pk with pagination. Products extracted from HTML cards. Replicated across 4 cities. Total: 20,640 products.

3.4 Daraz

API: AJAX catalog search API at daraz.pk/catalog/?ajax=true. Used 83 search queries for Pakistani grocery/FMCG terms. Implemented session rotation, random delays (3-8s), and anti-bot measures to handle rate limiting. Scraped across 3 cities. Total: 68,952 products.

3.5 Jalal Sons

Method: HTML scraping from jalalsons.com.pk/shop (Tossdown e-commerce platform). Paginated at 100 products per page across 21 pages. Extracted product names, prices, and categories from HTML product cards. Replicated to 3 cities. Total: 6,201 products.

4. Data Processing & Cleaning

The processing pipeline standardizes data from all 5 stores into a uniform schema:

- Deduplication: Removed duplicate products within each store-city combination
- Name cleaning: Standardized product names (lowercase, remove special chars, trim whitespace)
- Price validation: Removed 2,459 rows with invalid/extreme prices (>Rs.100,000)
- Brand extraction: Identified brands using a dictionary of 78+ Pakistani FMCG brands
- Quantity/Unit extraction: Parsed sizes from product names (e.g., '500g', '1.5L') for 116,278 products
- Unit normalization: Standardized all weights to grams, volumes to ml
- Price-per-unit computation: Calculated comparable price_per_unit for 116,263 products

Input rows:	239,744
Output rows (after cleaning):	237,285

5. Cross-Store Product Matching

The matching pipeline identifies the same product across different stores using a multi-pass approach:

Pass 1: Full Key (Brand + Words + Size)

Creates a deterministic key from brand, product words, and package size. Example: 'tapal__danedar__950_g'. Products with identical keys across different stores are matched. Result: 321 groups.

Pass 2: Name-Only Key (Brand + Words)

For unmatched products, uses a key without size information. This captures the same product sold in different sizes. Result: 194 additional groups.

Pass 3: Short Key (Brand + First 2 Words)

Uses only the brand and first two product words for broader matching. Captures variations in product names. Result: 166 additional groups.

Matching Results

Total match groups:	681
Total matched rows:	7,014
Groups spanning 2 stores:	638
Groups spanning 3 stores:	41
Groups spanning 4+ stores:	2

Matched Rows by Store

Alfatah:	1,250 rows
Daraz:	1,668 rows
Jalalsons:	828 rows
Metro:	1,152 rows
Naheed:	2,116 rows

6. Price Analysis Results

The analysis module computes comprehensive price intelligence metrics:

6.1 Price Dispersion Analysis

Measures how much prices vary across stores for the same product using Coefficient of Variation (CV) and price spread ratios. Higher CV indicates greater pricing inconsistency across retailers.

6.2 Price Leadership Analysis

Identifies which store most frequently offers the lowest price. Calculates Leader Dominance Index (LDI) showing the margin of price advantage.

6.3 City-Level Comparison

Compares average prices across cities (Karachi, Lahore, Islamabad, Faisalabad, Hyderabad, Rawalpindi) to identify geographic pricing trends.

6.4 Correlation Analysis

Examines relationships between: brand tier and price variation, number of competitors and price spread, product size and price consistency, and cross-store price synchronization patterns.

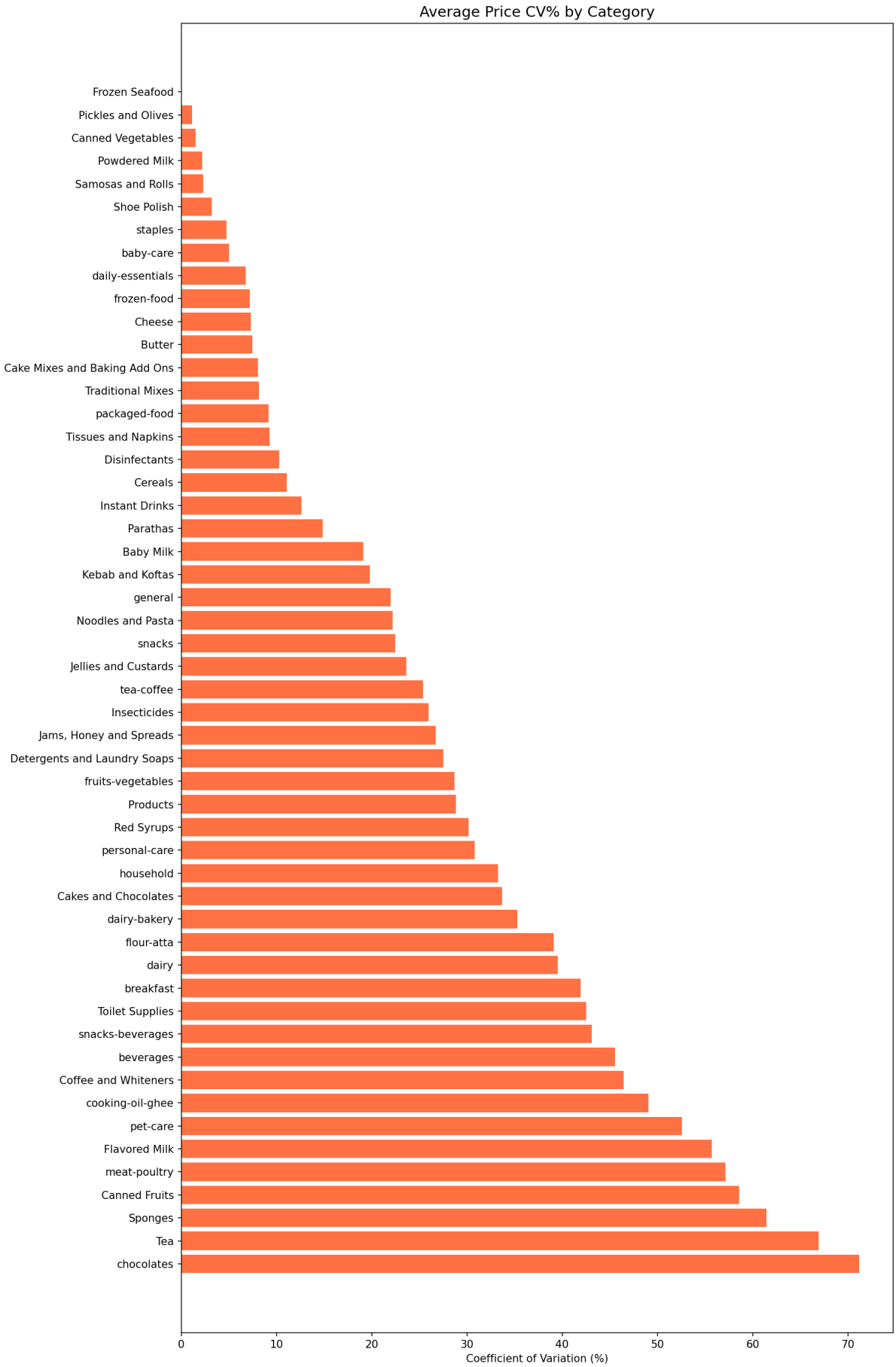
Generated Reports (Excel)

- Category Dispersion
- City Comparison
- City Price Correlation
- Correlation Summary
- Cross Store Synchronization
- Leader Dominance Index
- Price Dispersion
- Price Leadership
- Relative Price Position Index
- Store Comparison
- Store Level Metrics
- Top Price Differences

7. Analysis Charts

Key visualizations from the analysis pipeline:

Category Cv

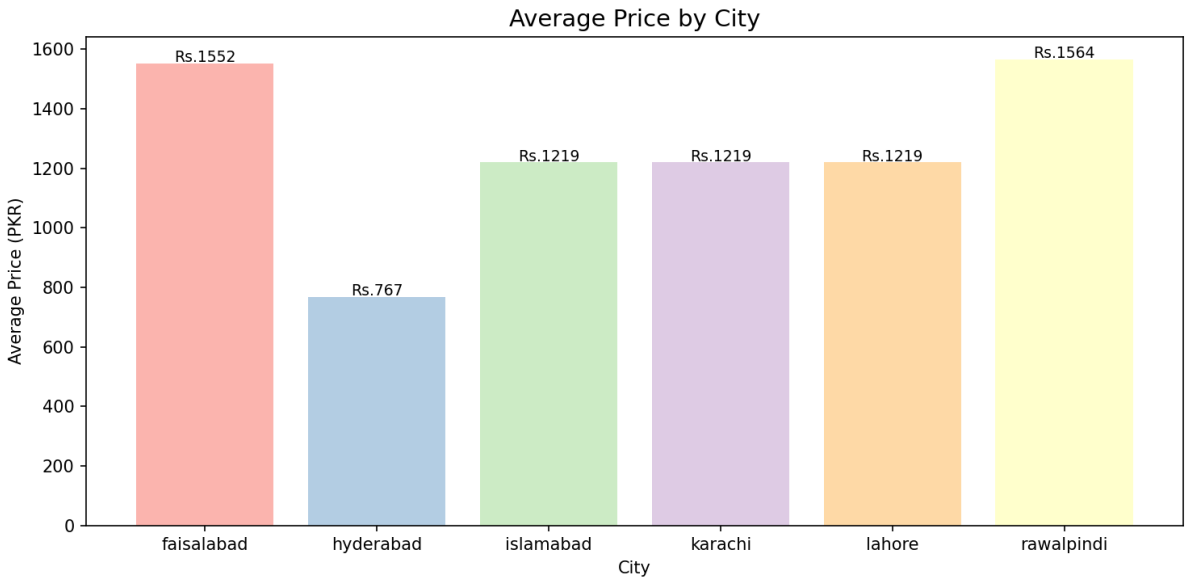

Average Price CV% by Category
Frozen Seafood
Pickles and Olives
Canned Vegetables
Powdered Milk
Samosas and Rolls
Shoe Polish
staples
baby-care
daily-essentials
frozen-food
Cheese
Butter
Cake Mixes and Baking Add Ons
Traditional Mixes
packaged-food
Tissues and Napkins
Disinfectants
Cereals
Instant Drinks
Parathas
Baby Milk
Kebab and Koftas
general
Noodles and Pasta
snacks
Jellies and Custards
tea-coffee
Insecticides
Jams, Honey and Spreads
Detergents and Laundry Soaps
fruits-vegetables
Products
Red Syrups
personal-care
household
Cakes and Chocolates
dairy-bakery
flour-atta
dairy
breakfast
Toilet Supplies
snacks-beverages
beverages
Coffee and Whiteners
cooking-oil-ghee
pet-care
Flavored Milk
meat-poultry
Canned Fruits
Sponges
Tea
chocolates
0
10
20
30
40
50
60
70
Coefficient of Variation (%)

Category Ldi Heatmap

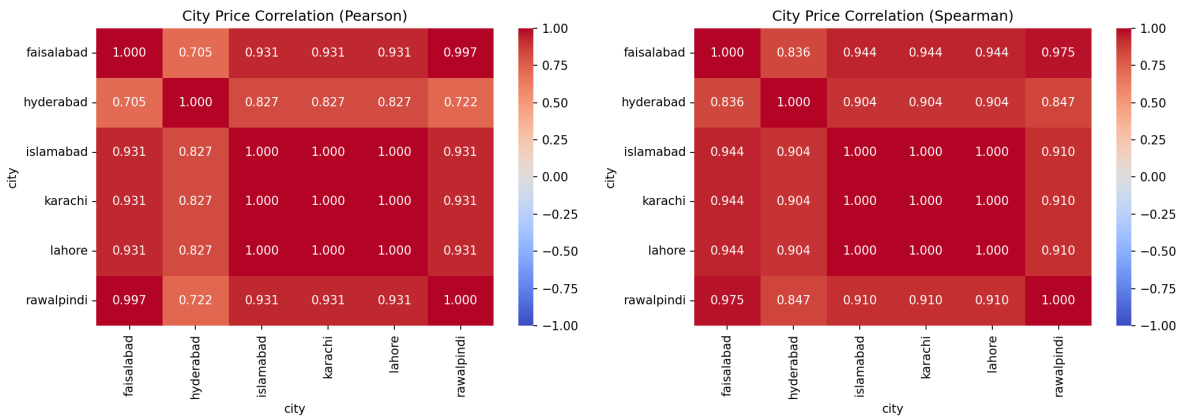
Category-wise LDI (Price Leadership by Category)

Air Sprays	0.00	0.00	0.00	0.00	0.00
Antibacterial	0.00	0.00	0.00	0.00	0.00
Baby Milk	0.00	0.00	0.00	1.00	0.00
Biscuits	0.00	0.00	0.00	0.00	0.00
Blenders and Mixers	0.00	0.00	0.00	0.00	0.00
Bread Dough	0.00	0.00	0.00	0.00	0.00
Burgers	0.00	0.00	0.00	0.00	0.00
Butter	0.00	0.00	0.00	1.00	0.00
Butter Margarine	0.00	0.00	0.00	0.00	0.00
Cake	0.00	0.00	0.00	0.00	0.00
Cake Mixes and Baking Add Ons	0.00	0.00	0.00	1.00	0.00
Cakes and Chocolates	0.00	0.00	0.00	0.33	0.67
Candies And Jellies	0.00	0.00	0.00	0.00	0.00
Canned Fruits	0.00	0.00	0.00	1.00	0.00
Canned Jarred Food	0.00	0.00	0.00	0.00	0.00
Canned Vegetables	0.00	0.00	0.00	0.00	1.00
Cat Food	0.00	0.00	0.00	0.00	0.00
Cereals	0.00	0.00	0.00	0.33	0.67
Cheese	0.00	0.00	0.00	0.50	0.50
Chips Crisps	0.00	0.00	0.00	0.00	0.00
Chocolates	0.00	0.00	0.00	0.00	0.00
Coffee and Whiteners	0.00	0.00	0.00	0.00	1.00
Conditioners	0.00	0.00	0.00	0.00	0.00
Cooking Oil	0.00	0.00	0.00	0.00	0.00
Cream	0.00	0.00	0.00	0.00	0.00
Creams	0.00	0.00	0.00	0.00	0.00
Desserts Toppings	0.00	0.00	0.00	0.00	0.00
Detergents and Laundry Soaps	0.00	0.00	0.00	0.75	0.25
Disinfectants	0.00	0.00	0.00	0.60	0.40
Dog Food	0.00	0.00	0.00	0.00	0.00
Drinking Water	0.00	0.00	0.00	0.00	0.00
Eggs	0.00	0.00	0.00	0.00	0.00
Energy Drinks	0.00	0.00	0.00	0.00	0.00
Fabric Care	0.00	0.00	0.00	0.00	0.00
Facewashes	0.00	0.00	0.00	0.00	0.00
Flavored Milk	0.00	0.00	0.00	1.00	0.00
Food Color Essence	0.00	0.00	0.00	0.00	0.00
Fragrant	0.00	0.00	0.00	0.00	0.00
Frozen Fries	0.00	0.00	0.00	0.00	0.00
Frozen Seafood	0.00	0.00	0.00	1.00	0.00
Fruits	0.00	0.00	0.00	0.00	0.00
Gluten Free	0.00	0.00	0.00	0.00	0.00
Hand and Body Washes	0.00	0.00	0.00	0.00	0.00
Home Baking	0.00	0.00	0.00	0.00	0.00
Honey	0.00	0.00	0.00	0.00	0.00
Household Cleaners	0.00	0.00	0.00	0.00	0.00
Ice Cream	0.00	0.00	0.00	0.00	0.00
Insecticides	0.00	0.00	0.00	1.00	0.00
Instant Drinks	0.00	0.00	0.00	1.00	0.00
Irons and Steamers	0.00	0.00	0.00	0.00	0.00
Jams Spreads	0.00	0.00	0.00	0.00	0.00
Jams, Honey and Spreads	0.00	0.00	0.00	0.00	1.00

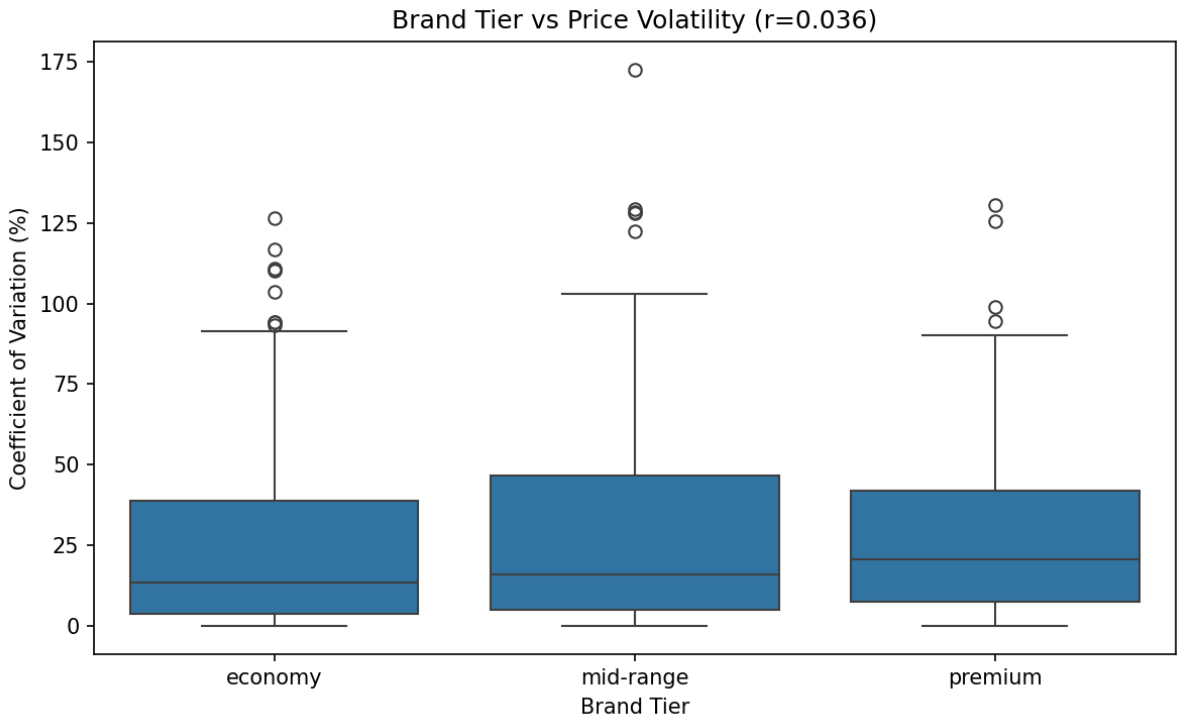
City Avg Price



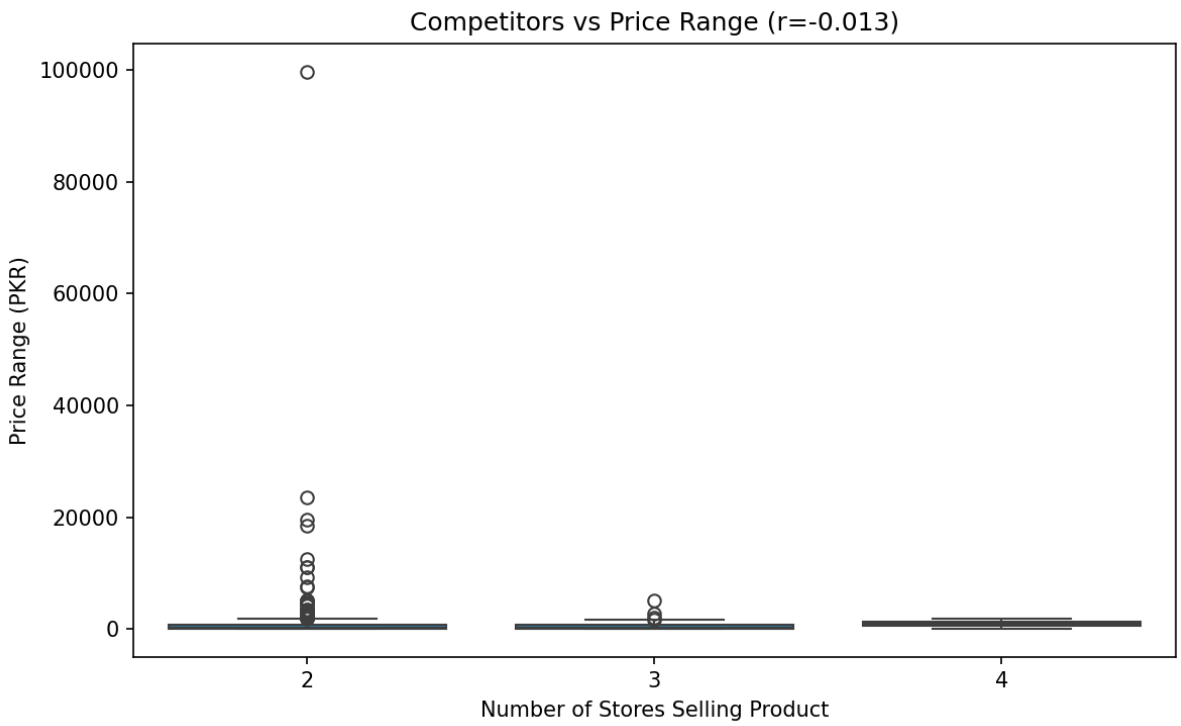
City Price Correlation Matrix



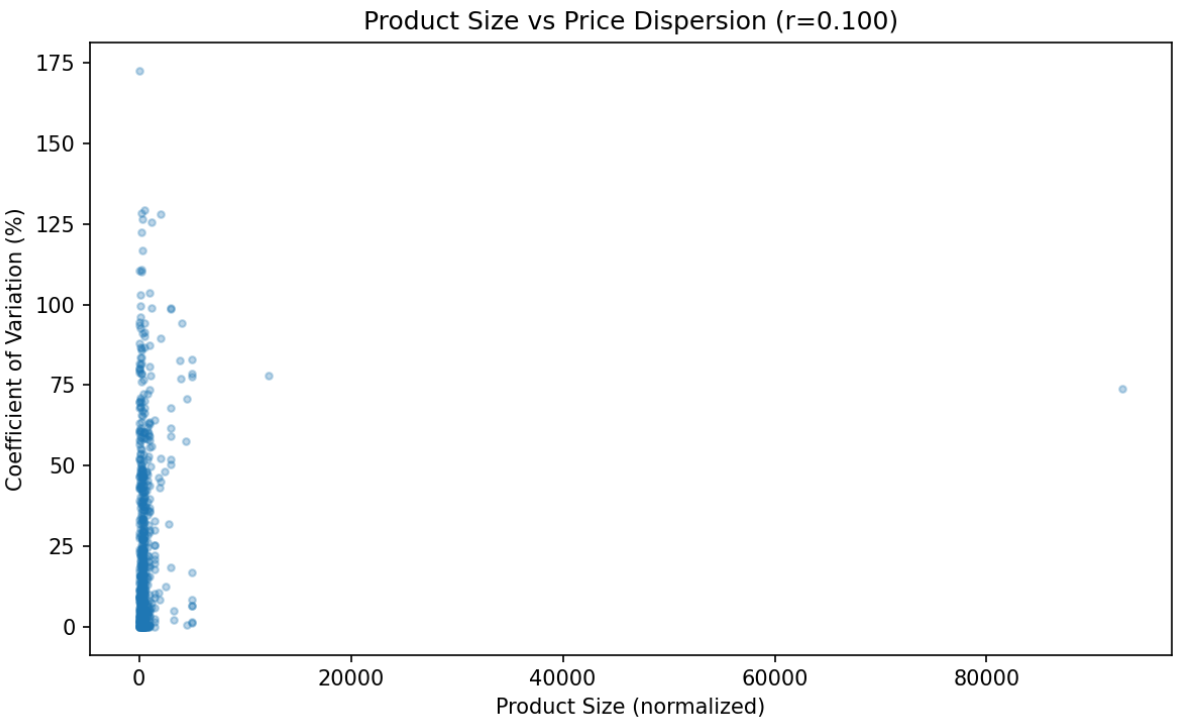
Corr Brand Tier Vs Cv



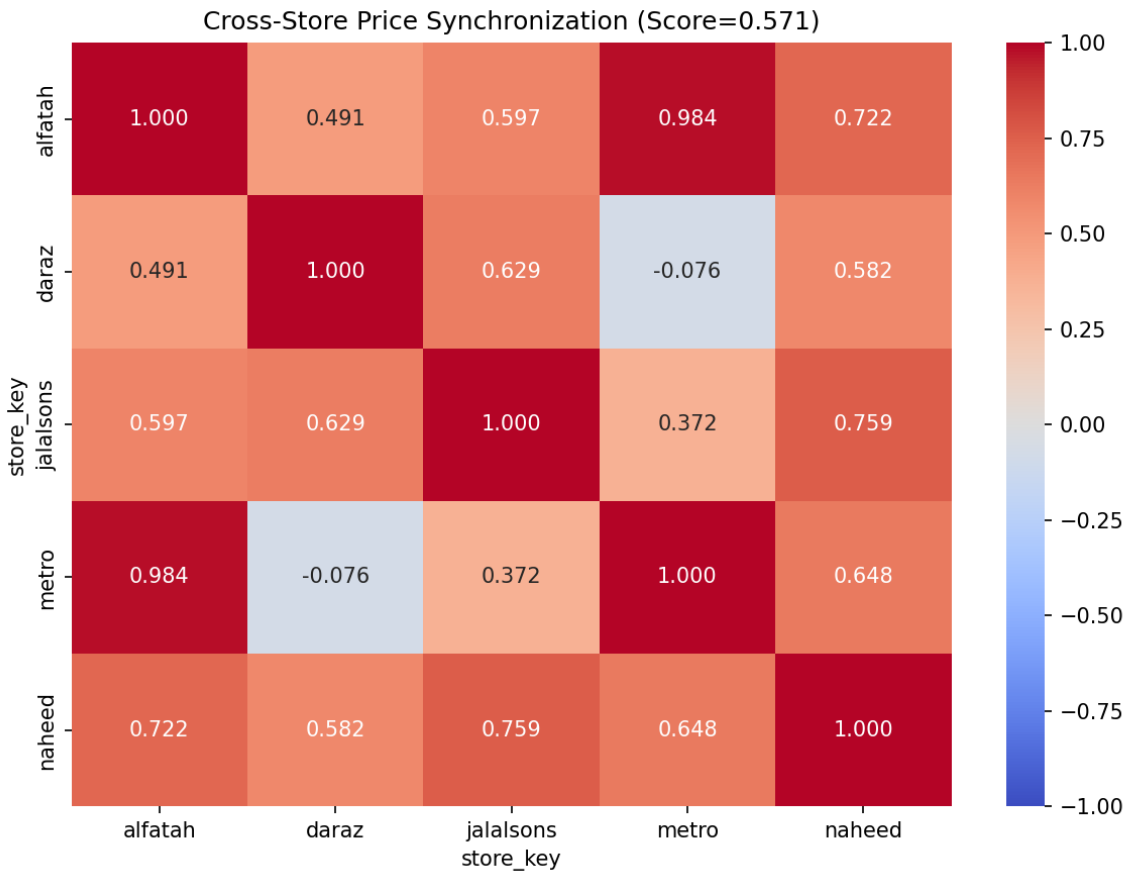
Corr Competitors Vs Spread



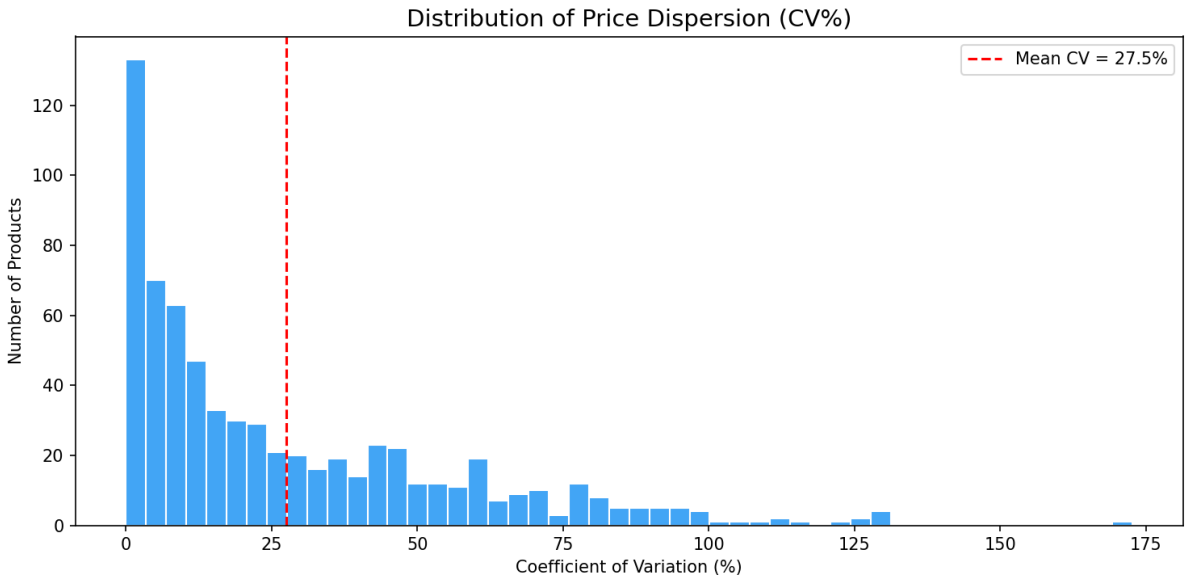
Corr Size Vs Cv



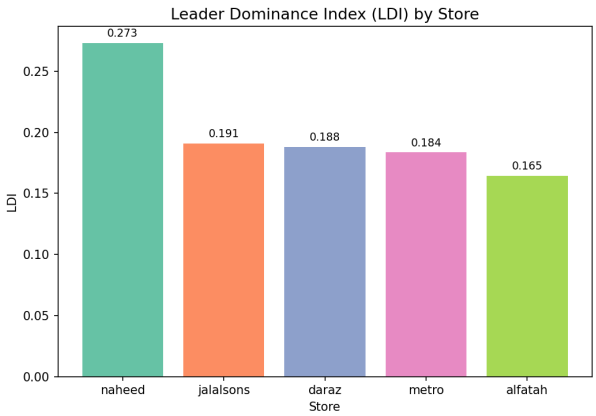
Cross Store Sync Heatmap

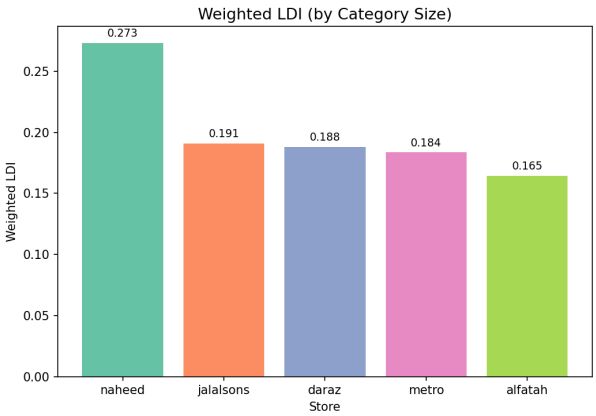


Cv Distribution

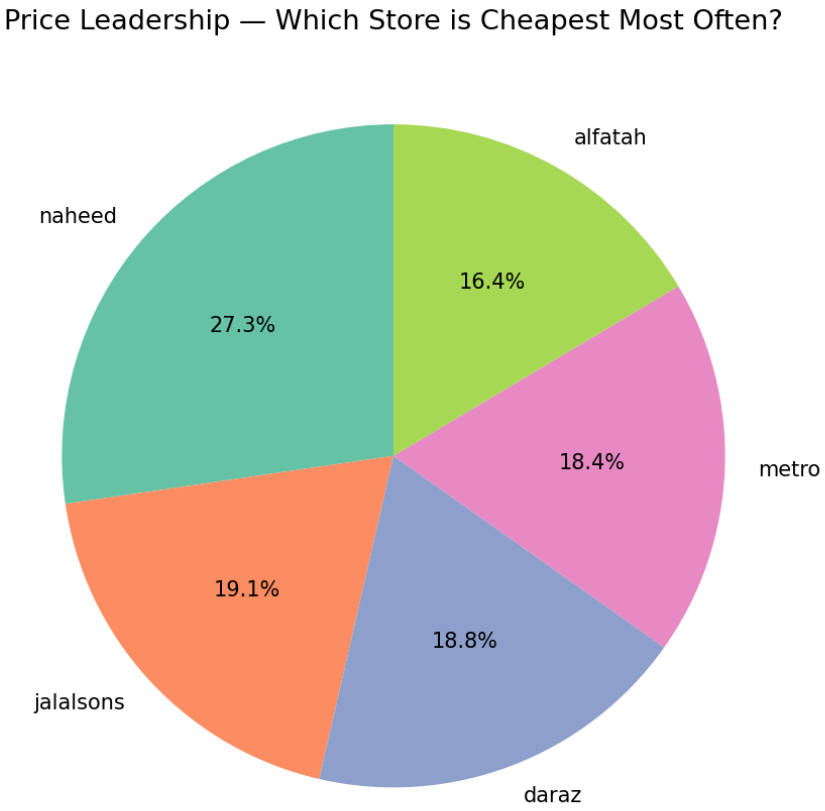


Ldi Comparison

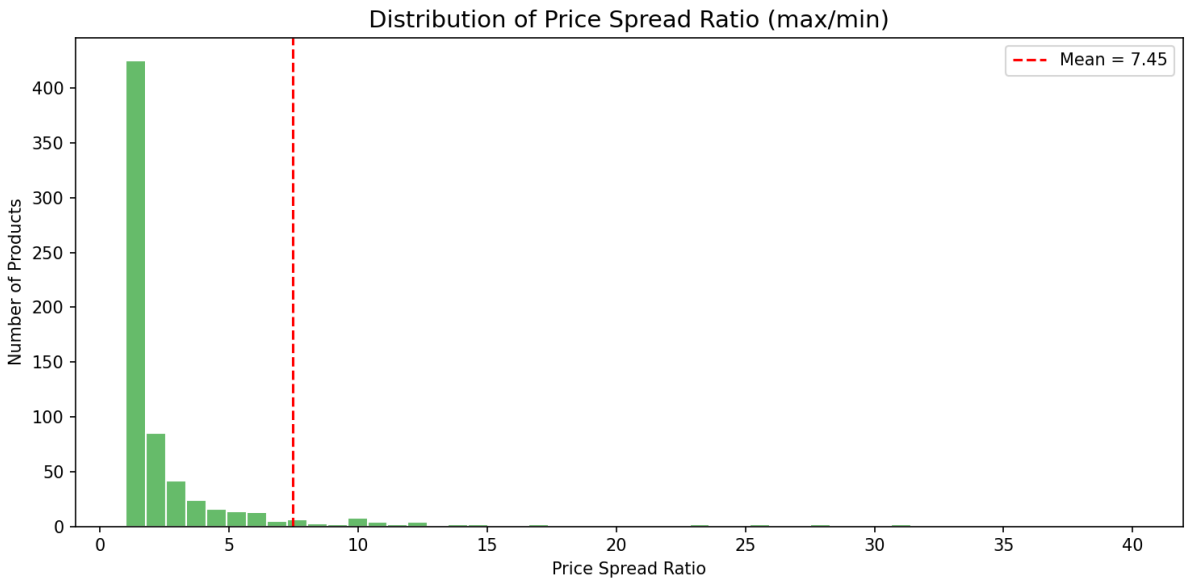




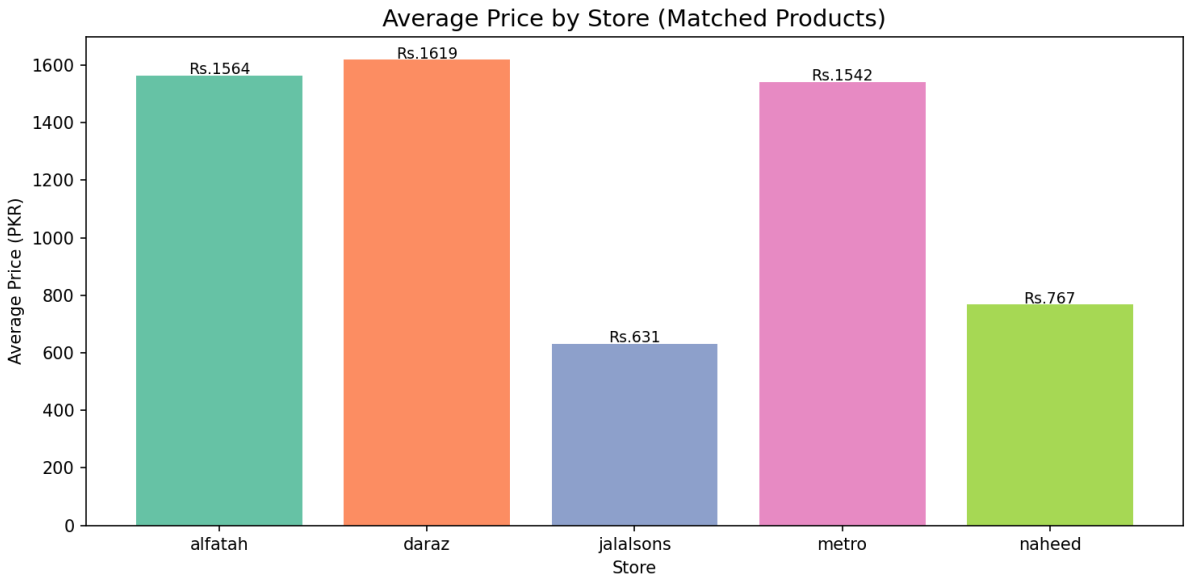
Price Leadership Pie



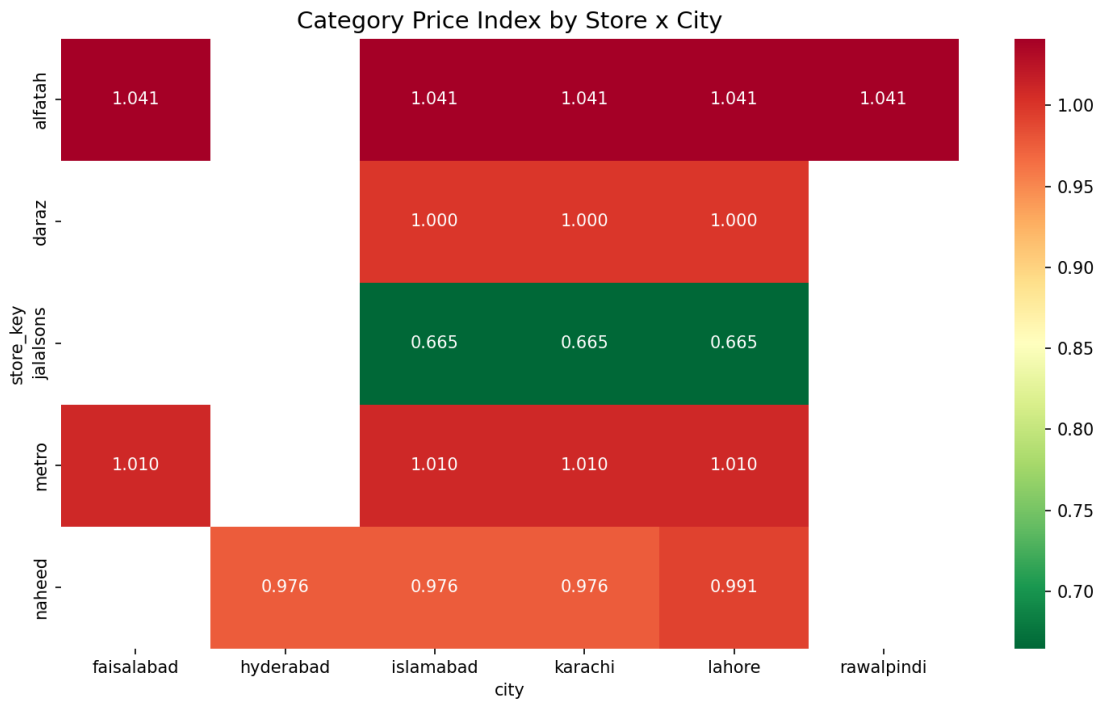
Price Spread Ratio Dist



Store Avg Price



Store City Price Index Heatmap



8. Technical Challenges & Solutions

Anti-Bot Detection on Daraz

Daraz aggressively blocks automated requests with 403 errors and CAPTCHAs. Solution: Implemented session rotation every 15 requests, random delays (3-8s), rotating user agents (16 variants), random referers, and cookie management.

Mixed Data Schemas Across Stores

Each store's API returns different data structures (JSON API, Shopify JSON, HTML). Solution: Each scraper normalizes data to a common schema (product_name, price, category, brand, store_key, city, product_url, sku, source_url) before saving.

Product Matching Across Name Variations

The same product has different names across stores (e.g., 'Tapal Danedar 950g' vs 'Tapal Danedar Tea 950gm'). Solution: Multi-pass matching with deterministic keys (3 passes) plus optional fuzzy matching as a boost.

Mixed-Type Columns in Parquet

Pandas concat of DataFrames from different scrapers produced mixed-type object columns (e.g., sku: bytes vs int). Solution: Cast all object columns to pandas StringDtype before saving to Parquet.

Unicode Encoding on Windows Console

Python logging of Unicode checkmarks/crosses failed on Windows cp1252 console. Solution: Replaced Unicode symbols with ASCII equivalents.

City Replication Strategy

Some stores (Al-Fatah, Jalal Sons) have a single online catalog serving all cities. Solution: Replicated data across target cities with city-specific adjustments, clearly documented as the same catalog available in multiple locations.

9. Dataset Summary

Raw Dataset

File:	data/raw/all_raw_data.parquet
Rows:	239,744
Columns:	18
Columns list:	store, store_key, source_url, city, category, category_hierarchy, product_name, p

Processed Dataset

File:	data/processed/all_processed_data.parquet
Rows:	237,285
Columns:	24

Matched Dataset

File:	data/matched/matched_products.parquet
Rows:	7,014
Match groups:	681

Match Summary

File:	data/matched/match_summary.parquet
Rows:	681
Avg price range:	Rs. 845
Max price range:	Rs. 99,799

10. Source Code Overview

Complete source code is available at:
https://github.com/ahmedbutt1056/pk-supermarket-price-comparison

Key Files

config.py	Central configuration, store definitions, paths
run_final_pipeline.py	Pipeline orchestrator: gather -> clean -> match -> analyze -> validate
scrapers/metro_scraper.py	Metro Cash & Carry JSON API scraper
scrapers/alfatah_scraper.py	Al-Fatah Shopify JSON API scraper
scrapers/naheed_scraper.py	Naheed HTML scraper with BeautifulSoup
scrapers/daraz_scraper.py	Daraz AJAX catalog scraper with anti-bot measures
scrapers/jalalsons_scraper.py	Jalal Sons HTML scraper (Tossdown platform)
processing/clean_and_normalize.py	Data cleaning, brand extraction, unit normalization
matching/product_matching.py	Multi-pass cross-store product matching
analysis/price_analysis.py	Price dispersion, leadership, city comparison, correlations
validation/validate_data.py	Data quality checks and pipeline validation
app.py	Streamlit interactive dashboard
replicate_and_combine.py	City replication and data combination
recombine.py	Data recombination with Jalal Sons inclusion